

# Kouds HALITIM

## PhD student in control of HPC systems

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## SKILLS

|                   |                                                                                                                                                                                                               |
|-------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Research          | Control theory for computing systems, Feedback control in distributed infrastructures, Thermal-aware resource management, Congestion mitigation in storage systems, Hybrid control and scheduling frameworks. |
| Programming       | C, C++, Python, MATLAB, R                                                                                                                                                                                     |
| Databases         | MySQL                                                                                                                                                                                                         |
| Tools & DevOps    | Git, Docker, Make, Nix, VS Code, oar                                                                                                                                                                          |
| Operating Systems | Windows 11, Linux Ubuntu                                                                                                                                                                                      |

## WORK EXPERIENCE

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|--------------------------|--------------------------------------------------------------------------------------------------------------------------|
| Present<br>Nov. 2023     | <b>PhD Candidate in Control of HPC systems, INRIA</b><br>Ctrl-A Team, Inria, LIG, Univ. Grenoble Alpes, Grenoble, France |
| Sept. 2023<br>March 2023 | <b>Research Intern, INRIA</b><br>Spirals Team, Inria, Lille, France                                                      |
| Jan. 2023<br>Dec. 2022   | <b>Industrial Automation Intern, Sonatrach</b><br>Sonatrach, Skikda, Algeria                                             |
| March 2022<br>April 2022 | <b>Embedded Systems Intern, Brandt</b><br>Brandt, Sétif, Algeria                                                         |
| April 2021<br>March 2021 | <b>Instrumentation Intern, IRIS</b><br>IRIS, Sétif, Algeria                                                              |

## EDUCATION

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| November 2026<br>(Expected) | <b>PhD in Computer Science, INRIA</b><br>Ctrl-A Team, Inria, LIG, Univ. Grenoble Alpes, Grenoble, France. "Efficient task hybridization in heterogeneous computing : a practical combination of control and scheduling theories."<br>Supervised by : Eric Rutten, Raphael Bleuse, and Bogdan Robu. <ul style="list-style-type: none"><li>➤ Developed models for large-scale distributed systems (HPC) to account for system dynamics, resource constraints, and thermal uncertainties.</li><li>➤ Synthesized and implemented controllers to regulate trade-offs between file system congestion, resource harvesting and power consumption.</li><li>➤ Designed a Stochastic Model Predictive Control framework to optimize task injection under high uncertainty and variable workload conditions.</li><li>➤ Formulated I/O interference as a control problem and implemented regulation mechanisms for multi-client environments to ensure performance isolation.</li><li>➤ Built large-scale experimental testbeds and automated data collection to validate theoretical models against state-of-the-art heuristics.</li><li>➤ Conducted robustness analyses and comparative evaluations to assess the stability, performance and scalability of implemented controllers.</li><li>➤ Designed a hierarchical regulation framework for processor power capping that ensures energy budget compliance across variable system dimensions and hardware configurations.</li></ul> |
| September 2023              | <b>Master's and Engineering Degree, ENP</b><br>National Polytechnic School, Algiers, Algeria.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| September 2018              | <b>High School Diploma in Mathematics and Physics, with Honors</b><br>Fatima Zahra High School, Sétif, Algeria                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |

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| June 2026<br>September 2024 | <b>Discrete-Time Control Systems, Lab Sessions - 18 hours</b> <ul style="list-style-type: none"> <li>➤ Developed comprehensive lab curricula and source code for a digital motion control lab, integrating micro-controllers with real-time data processing.</li> <li>➤ Mentored diverse student groups (15-20) by applying theoretical control concepts to real-world experimental scenarios.</li> <li>➤ Collaborated proactively with the teaching team to improve educational materials.</li> </ul>                                                                                                                                                                                                                                                                               |
| June 2026<br>September 2024 | <b>Continuous-Time Control Systems, Lab Sessions - 59 hours</b> <ul style="list-style-type: none"> <li>➤ Contributed to the drafting and updating of laboratory workbooks.</li> <li>➤ Created digital supporting resources on MATLAB/Simulink to facilitate the adoption of calculation and simulation tools.</li> <li>➤ Commissioned, tested, and configured specific testbeds (PID Controllers and Rotating Machines) and experimental rigs (DC motors, sensors).</li> <li>➤ Led the full control design process : modeling, experimental identification, and PID synthesis to meet performance criteria.</li> </ul>                                                                                                                                                               |
| April 2026                  | <b>Software Engineering Project, Project Tutor - 5.5 hours</b> <ul style="list-style-type: none"> <li>➤ Supervised student teams in the full-cycle development of strategy game software.</li> <li>➤ Mentored groups on Agile methodologies : task planning, workload distribution, and conflict resolution.</li> <li>➤ Provided technical oversight on the implementation of Minimax algorithms and heuristic optimization.</li> <li>➤ Guided the design and validation of Graphical User Interfaces (GUI) using user-centered testing.</li> <li>➤ Evaluated technical deliverables, including code quality, architectural robustness, and documentation.</li> <li>➤ Acted as a member of the final defense jury to assess project outcomes and technical presentations.</li> </ul> |

 SUPERVISION

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| 2026 | <b>Doaa Elnoamany</b> : “ <i>Control of uncertain MIMO systems with probabilistic constraints...</i> ”<br>Master of Science (M.Sc.) MiSCIT, Université Grenoble Alpes, Grenoble, France.<br>Co-encadrement : Bogdan Robu and Sophie Cerf                                       |
| 2025 | <b>Mohamed Abdeljalil Maziz</b> : “ <i>LPV control for dynamic power capping in High-Performance Computing (HPC) under mixed workloads</i> ”<br>Master of Science (M.Sc.) MiSCIT, Université Grenoble Alpes, Grenoble, France.<br>Co-encadrement : Bogdan Robu and Sophie Cerf |
| 2025 | <b>Mahmoud Abdo</b> : “ <i>Model Predictive Control (MPC) approach for resource scavenging in High-Performance Computing (HPC)</i> ”<br>Master of Science (M.Sc.) MiSCIT, Université Grenoble Alpes, Grenoble, France.<br>Co-encadrement : Bogdan Robu and Sophie Cerf         |
| 2024 | <b>Jawad Hasnain</b> : “ <i>Extremum Seeking Control (ESC) for first-order systems with various objective functions</i> ”<br>Master's (M1) in Electrical Engineering and Control Systems, Université Grenoble Alpes, Grenoble, France.<br>Co-encadrement : Bogdan Robu         |

 PUBLICATIONS

## International Conferences

- [C1] **K. Halitim**, T. Collignon, S. Cerf, B. Robu, E. Rutten, R. Bleuse, “*Event-Triggered Adaptive PI Control for Mitigating Congestion in Shared Storage Systems*”, in ECC 2026 - European Control Conference, Reykjavik, Iceland, 2026. (Accepted).
- [C2] **K. Halitim**, S. Cerf, B. Robu, and É. Rutten, “*Robust Cascade Control for Variable Dimension Systems - Application to processors' power capping*”, in ECC 2025 - European Control Conference, Thessaloniki, Greece, 2025. Available : <https://hal.science/hal-05117585v1>
- [C3] **K. Halitim**, B. Robu, S. Cerf, R. Bleuse, E. Rutten, “*Stochastic Model Predictive Control with Direct Feedforward Compensation*”, in ICSTCC 2025 - 29th International Conference on System Theory, Control and Computing, Cluj-Napoca, Romania, 2025. Available : <https://hal.science/hal-05210549v1>

[C4] T. Collignon, **K. Halitim**, R. Bleuse, S. Cerf, B. Robu, É. Rutten, L. Seinturier, and A. van Kempen, "Mitigating Shared Storage Congestion Using Control Theory", in UCC 2025 - IEEE/ACM International Conference on Utility and Cloud Computing, Nantes, France, Dec. 2025. Available : <https://cnrs.hal.science/LIG/hal-05368563v1>

National Conferences

[N1] T. Collignon, **K. Halitim**, R. Bleuse, S. Cerf, B. Robu, É. Rutten, L. Seinturier, and A. van Kempen, "Coarse-grain Congestion Regulation Using Control Theory", in COMPAS 2025 - Conférence en Parallélisme, Architecture et Système, Bordeaux, France, 2025. Available : <https://hal.science/hal-05075480v1>

Theses

[T1] **Kouids Halitim**. "Enhancing Efficiency through Control Theory in Compute-Intensive Applications," Master's Thesis, 2023. Available : <https://inria.hal.science/hal-04357812>

Posters

[P1] T. Collignon, **K. Halitim**, R. Bleuse, S. Cerf, B. Robu, É. Rutten, L. Seinturier, and A. van Kempen, "Temperature control of computing sites through the injection of useful and non-invasive tasks", Greendays 2026, March 2026, Sophia, France. Available : <https://inria.hal.science/hal-05551250v1>

[P2] **Kouids Halitim**. "Task hybridization in heterogeneous computing : a combination of control and scheduling theories", "PAD Days" GIPSA, December 2025, GRENOBLE (38000), France.

[P3] **Kouids Halitim**. "Task hybridization in heterogeneous computing : a combination of control and scheduling theories", "PhD Madness" LIG, May 2025, GRENOBLE (38000), France. Available : <https://inria.hal.science/hal-05108245v1>

 SCIENTIFIC RESPONSIBILITIES

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|------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 2025 | Reviewer for the European Control Conference (ECC 2025) and the International Conference on Utility and Cloud Computing (UCC 2025)                                |
| 2025 | Session Chair for the "Control Applications" session at the European Control Conference (ECC 2025)                                                                |
| 2026 | Session Chair for the "Heterogeneous and reconfigurable architectures for the future of computing : Project Talks" session at the 18 <sup>th</sup> JLESC workshop |

 LANGUAGES

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|-----------|----------|
| French :  | Fluent   |
| English : | Fluent   |
| Arabic :  | Native   |
| German :  | Beginner |